From 191 to 25 covariates: a parsimonious and interpretable linear model for the Italian Fragility Composite Index

Applied Statistics Project — Extension Report

Abstract

This report documents an iteration of the IFC linear benchmark made in response to instructor feedback. The first benchmark, with 191 covariates selected by LASSO, reached an out-of-sample R^2 of 0.825 but was too dense to discuss substantively. We therefore derived a parsimonious model in three explicit steps: iterative removal of multicollinearity via Variance Inflation Factor, top-N ranking by standardised effect, and a significance filter that replaces any non-significant coefficient with the next candidate from the ranking. Following an explicit substantive request, the geographic surface area of the municipality is excluded upfront, because it is a static descriptor rather than a fragility driver once the rest of the model is expressed in per-capita densities. The resulting model has 25 covariates, all significant at $p < 10^{-6}$, with a maximum VIF of 6.4. Its out-of-sample R^2 is 0.784, a controlled loss of 0.041 compared with the 191-covariate benchmark. The linear mixed-model extension with a region/province nested random intercept recovers almost all of that loss, reaching $R^2 = 0.818$, only 0.007 below the original full benchmark. A Moran's I on the residuals still detects significant positive spatial autocorrelation ($I = 0.234$, $p < 0.001$), confirming that the spatial structure is intrinsic to Italian municipal fragility and is not an artefact of the covariate set. Additional diagnostics on calibration, prediction uncertainty and residual normality complete the assessment and are discussed in dedicated sections.

Why this extension was necessary

After presenting the first benchmark, the instructor raised four concerns:

- **too many covariates**: 191 is unreadable in a poster or a single slide; 10–20 is the working range an audience can follow;
- **multicollinearity**: the first report only had a qualitative remark on $PM_{10}/PM_{2.5}$; a systematic Variance Inflation Factor analysis would tell us how many of the 191 are independent informative predictors;
- **ridge vs lasso**: an Elastic Net comparison across mixing parameters would tell us whether the choice of regulariser drives the result;
- **surface area**: the variable `Superficie_totale_Kmq` is a static feature of a municipality and should not appear in a fragility model where every other count has already been turned into a per-capita density.

This extension addresses all four. The numerical comparison with the original 191-covariate run is explicit throughout.

1 Model specification

We use four nested model variants. Throughout, i indexes the municipality $\times$ year observation and j indexes the covariate. Predictors X_{ji} are standardised so that $\hat{\beta}_j$ is the change in IFC associated with a one-standard-deviation shift of the j-th covariate.

Parsimonious linear model (LM). The baseline is the ordinary linear model with 25 fixed covariates:

$$\text{IFC}_i \;=\; \beta_0 + \sum_{j=1}^{25} \beta_j X_{ji} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2). \tag{1}$$

Region-level random intercept (M_{geo}). Adds a regional baseline:

$$\text{IFC}_i \;=\; \beta_0 + \sum_{j=1}^{25} \beta_j X_{ji} + u_{r(i)} + \varepsilon_i, \qquad u_r \sim \mathcal{N}(0, \sigma_R^2), \tag{2}$$

where $r(i) \in \{1, \dots, 20\}$ is the Italian region of municipality i. The term $u_{r(i)}$ is the average residual fragility of region r that the fixed effects cannot explain.

Region / province nested random intercept ($M_{\text{geo,nested}}$). Adds a second nested layer at the province level:

$$\text{IFC}_i \;=\; \beta_0 + \sum_{j=1}^{25} \beta_j X_{ji} + u_{r(i)} + v_{p(i)|r(i)} + \varepsilon_i, \qquad v_{p|r} \sim \mathcal{N}(0, \sigma_P^2), \tag{3}$$

with $p(i)$ one of 107 provinces. Each province has its own residual fragility on top of the regional one.

GRINS macroclass random intercept (M_{tax}). Replaces the geographic hierarchy with the GRINS taxonomy:

$$\text{IFC}_i \;=\; \beta_0 + \sum_{j=1}^{25} \beta_j X_{ji} + t_{m(i)} + \varepsilon_i, \qquad t_m \sim \mathcal{N}(0, \sigma_T^2), \tag{4}$$

where $m(i) \in \{\text{Internal Italy}, \text{Middle Italy}, \text{Metropolitan Italy}\}$ is the GRINS macroclass.

Region + macroclass crossed ($M_{\text{geo+tax}}$). Combines region and macroclass as two crossed random intercepts. This tests whether the taxonomy adds anything beyond geography once both are in the same model:

$$\text{IFC}_i \;=\; \beta_0 + \sum_{j=1}^{25} \beta_j X_{ji} + u_{r(i)} + t_{m(i)} + \varepsilon_i. \tag{5}$$

2 Method

2.1 Variance Inflation Factor pruning

For a linear model with p predictors,

$$\text{VIF}_j \;=\; \frac{1}{1 - R_j^2}, \quad R_j^2 = R^2(X_j \sim X_{-j}), \tag{6}$$

is the share of variance of X_j that the other predictors already explain. VIF $= 1$ corresponds to perfect independence; VIF > 10 is the conventional flag for serious collinearity.

Starting from the 191 LASSO-selected predictors (with surface area excluded upfront), we iteratively drop the covariate with the largest VIF and refit. The procedure stops after **34** iterations when the maximum VIF falls below 10. The worst offenders have VIFs above one hundred (e.g. `mef_tot_classi_amm`, VIF = 159). The complete drop log is in `outputs/parsimonious_model/vif_drop_log.c`

2.2 Ranking and selection by standardised effect

On the 157 VIF-clean predictors we fit one linear model with standardised predictors and rank them by $|\hat{\beta}_j|$. Standardisation makes coefficients directly comparable across heterogeneous variables. The top $n = 25$ form the candidate set.

2.3 Significance filter

When the top-25 by $|\hat{\beta}|$ is refit, occasionally one of them is not statistically significant: its $|\hat{\beta}|$ is carried by some residual collinearity that the VIF threshold of 10 has not fully removed. The script automatically drops any coefficient with $p > 0.05$ and replaces it with the next candidate from the ranking, then refits and repeats until either all 25 are significant or the candidate pool is exhausted. In our run a single replacement was triggered.

2.4 Mixed-model extension and spatial diagnostic

We refit the four mixed-model variants of Section 1 on the new 25-covariate fixed-effects set, and recompute Moran's I on the residuals of $M_{\text{geo,nested}}$. Spatial weights are the five-nearest-neighbour symmetric graph used previously.

2.5 Elastic Net cross-check

To answer the "LASSO with a ridge penalty" question, Elastic Net is run on the 157 VIF-clean predictors for $\alpha \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$. $\alpha = 1$ is pure LASSO, $\alpha = 0$ would be pure Ridge (no selection).

3 Results

3.1 Parsimony vs accuracy

Figure 1 shows the trade-off curve from running panel-aware cross-validation at top-n sizes ranging from 5 to all 157 VIF-clean predictors.

3.2 The 25-covariate model and its substantive reading

After the significance filter, all 25 coefficients are significant at $p < 10^{-6}$ and the maximum VIF in the final model is 6.42. Table 1 lists them sorted by $|\hat{\beta}_j|$. All values are on the standardised scale: $\hat{\beta}_j$ is the expected change in IFC for a one-standard-deviation increase of X_j.

The top 5 covariates in detail

These are the five predictors with the strongest standardised effect on IFC. Each one is interpreted as "a one-standard-deviation increase of the covariate changes IFC by $\hat{\beta}_j$ points". Since the IFC scale is approximately 100 ± 4, a coefficient of ± 1 is a sizeable effect.

1. `reddito_26000_55000` ($\hat{\beta} = -1.03$). Mass of taxpayers declaring an income between 26 000 and 55 000, expressed per inhabitant. It is a proxy for the size of the local middle class. One standard deviation more of middle-class density reduces IFC by about one point: the local middle class is the single strongest protective driver of municipal fragility.

2. `numero_contribuenti` ($\hat{\beta} = -0.78$). Total number of declared taxpayers per inhabitant. *Why this is not redundant with the previous variable.* The two top predictors look similar at first sight, but they capture different things: `reddito_26000_55000` measures the *composition* of declared income (how many of your taxpayers fall in the middle-income bracket),

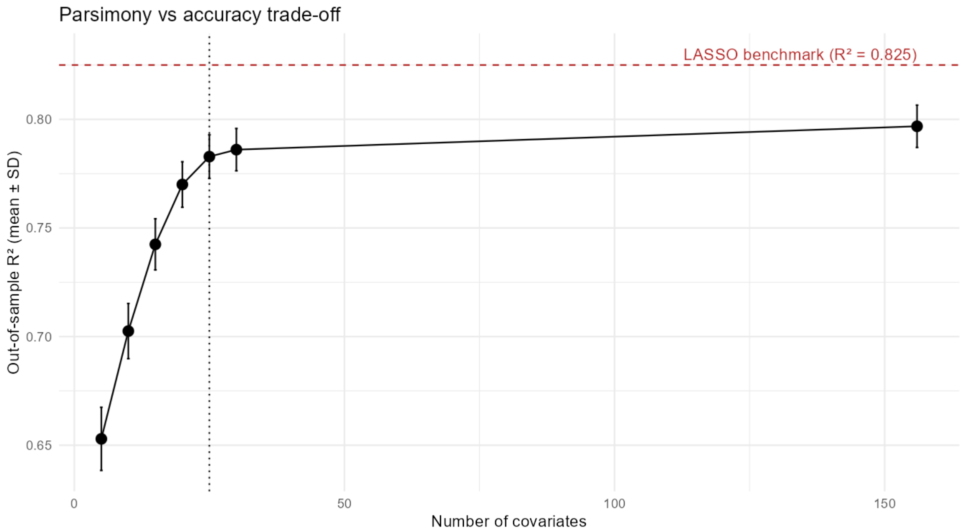

Figure 1: Out-of-sample R^2 vs the number of covariates kept from the VIF-clean ranking. Each point is the mean over 25 panel-aware cross-validation folds; the error bars are ± 1 standard deviation across folds. The red dashed line marks the R^2 of the original 191-covariate LASSO benchmark; the dotted vertical line marks our chosen $n_{\text{final}} = 25$. The elbow of the curve sits between 20 and 25: moving from 25 to 30 buys only $+0.004$ R^2, while moving from 25 down to 15 costs about 0.03.

Variable	$\hat{\beta}_{\text{std}}$	direction	substantive reading
`reddito_26000_55000`	-1.03	less fragile	middle-income taxpayer mass
`numero_contribuenti`	-0.78	less fragile	total taxpayer density
`mef_cls_0_10000_amm`	$+0.57$	more fragile	low-income declared mass
`reddito_pensione`	$+0.56$	more fragile	pension share of declared income
`asia_ul_f_w0_9`	-0.53	less fragile	small construction businesses density
`asia_ul_g_w0_9`	-0.49	less fragile	small retail/wholesale density
`asia_add_c_w0_9`	-0.48	less fragile	small manufacturing employment density
`frame_industria_VA_per_addetto`	-0.47	less fragile	industrial productivity
`asia_ul_m_w0_9`	-0.46	less fragile	small professional/scientific firm density
`frame_servizi_VA_per_addetto`	-0.45	less fragile	services-sector productivity
`frame_totale_retribuzione_su_VA`	-0.44	less fragile	wage share of value added
`lacc_mean`	$+0.43$	more fragile	mean local accessibility index
`asia_add_c_w10_49`	-0.38	less fragile	medium-small manufacturing employment
`bonus_trattamento`	-0.34	less fragile	wage-support beneficiaries density
`asia_add_s_w0_9`	-0.34	less fragile	small other-services employment
`mef_cls_75000_120000_freq`	-0.34	less fragile	upper-middle income taxpayer share
`mef_cls_55000_75000_freq`	-0.31	less fragile	upper-middle income taxpayer share
`employee_concentration`	-0.27	less fragile	employment concentration index
`asia_ul_q_w0_9`	-0.23	less fragile	small health-services density
`eta_media_media_donne`	-0.22	less fragile	mean age of female public employees
`asia_ul_i_w0_9`	-0.20	less fragile	small hospitality density
`Esercizi_complementari`	-0.18	less fragile	non-hotel tourism structures
`eta_uomini`	$+0.17$	more fragile	mean age of male population
`istr_occupazione_tempo_pieno_donne`	$+0.17$	more fragile	full-time female public employees
`no2_media_valori_annuali`	$+0.15$	more fragile	NO_2 annual mean

Table 1: Final 25 covariates after VIF pruning, top-25 ranking and significance filtering. All coefficients are significant at $p < 10^{-6}$, all in the standardised scale.

whereas `numero_contribuenti` measures the *level* of formal tax participation overall (how many people pay tax at all, regardless of bracket). Their sample correlation, after VIF pruning, is below 0.65, and a model that uses only one of them loses noticeable predictive power. Conceptually, `numero_contribuenti` is essentially an inverse proxy of the informal economy: the more people declare income, the smaller the share of underground activity. One SD more reduces IFC by 0.78 points.

3. `mef_cls_0_10000_amm` ($\hat{\beta} = +0.57$). Total amount declared in the €0–€10 000 income bracket, per inhabitant. The positive sign means that the more declared income is concentrated in the lowest bracket, the more fragile the municipality. It is the strongest direct proxy of economic distress.

4. `reddito_pensione` ($\hat{\beta} = +0.56$). Share of declared income coming from pensions, per inhabitant. Positive sign: municipalities whose income is heavily pension- based are more fragile. This is the demographic-aging channel: more pensioners, fewer active workers, less economic dynamism.

5. `asia_ul_f_w0_9` ($\hat{\beta} = -0.53$). Density of small local units (0–9 employees) in the construction sector (NACE F). Negative sign: a vibrant fabric of micro-construction firms is associated with lower fragility. It is one of several small-business indicators in the model (manufacturing, retail, professional services) that all point in the same direction.

The substantive narrative is clear and consistent: *fragility is driven by the balance between economic dynamism (taxpayer mass, small-business density, productivity) and economic distress (low-income mass, pension dependency).* Environmental indicators (NO_2) and demographic-aging indicators appear in the model but with smaller effects.

Notes on two technical names. Two variables in the table have non-self-explanatory names and deserve a short clarification.

- `lacc_mean` ($\hat{\beta} = +0.43$) is the mean of a *local accessibility* index pre-computed in the GRINS data set. The positive sign — more accessibility associated with higher predicted fragility — looks counterintuitive at first. The most likely explanation is that, after we have controlled for twenty-four other socio-economic variables, what remains in `lacc_mean` is a residual urbanisation/density signal that the rest of the model has not absorbed. The substantive interpretation should be treated with caution; the data dictionary of GRINS V3 should be consulted for the exact construction before strong claims are made on this coefficient.

- `employee_concentration` ($\hat{\beta} = -0.27$) is a Herfindahl-type concentration index of employment across sectors. We report the standardised coefficient as we estimated it; a strict substantive reading is again deferred to a check of the variable definition.

The remaining 23 variables have transparent names and effects, and form the basis of the discussion above.

3.3 Elastic Net cross-check

Table 2 reports the selection size and the cross-validated MSE for the five values of α on the VIF-clean set. Figure 2 shows the same information graphically.

3.4 Mixed-model extension

Table 3 reports the out-of-sample R^2 of every model variant on both covariate sets. Figure 3 visualises the same numbers as a bar chart.

α	# selected	λ_{1se}	CV MSE
0.10 (ridge-leaning)	124	0.108	3.512
0.25	111	0.060	3.526
0.50	114	0.028	3.520
0.75	88	0.028	3.546
1.00 (pure LASSO)	93	0.019	3.537

Table 2: Elastic Net comparison on the 157-covariate VIF-clean set.

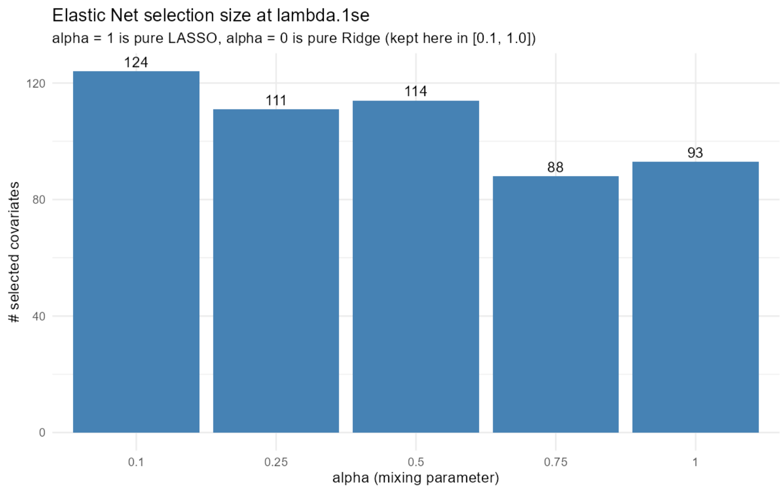

Figure 2: Number of covariates selected by Elastic Net at λ_{1se} for five values of the mixing parameter α. $\alpha = 1$ is pure LASSO; $\alpha = 0$ would be pure Ridge (without variable selection). Across the entire range, the cross-validated MSE varies by less than one percent (Table 2) and the selection sizes are bunched between 88 and 124. On a feature matrix already cleaned of strong collinearity, the choice of regulariser is therefore not critical: LASSO is a reasonable default.

Model	R^2 (191 cov, old)	R^2 (25 cov, new)	Δ
Linear model (no random effect)	0.826	0.784	−0.042
M_{tax}	0.829	0.784	−0.045
M_{geo}	0.846	0.815	−0.031
$M_{\text{geo+tax}}$	0.846	0.812	−0.034
$M_{\text{geo,nested}}$	0.853	**0.818**	−0.035

Table 3: Linear and mixed-model R^2 with the 191-covariate and the 25-covariate fixed-effects sets, evaluated by the same five-fold panel-aware cross-validation.

Macroclass is redundant with geography. $M_{\text{geo+tax}}$ is statistically indistinguishable from M_{geo}. Once region is in the model, the GRINS macroclass adds nothing. This is the formal evidence that the macro-taxonomy and the geographic macro-structure of Italy carry the same information.

Variance components and intra-class correlation. The best mixed model, $M_{\text{geo,nested}}$, decomposes the remaining variance of IFC after the 25 fixed effects into three pieces:

$$\widehat{\sigma}_{\text{region}} = 1.22, \qquad \widehat{\sigma}_{\text{province}|\text{region}} = 0.50, \qquad \widehat{\sigma}_{\text{residual}} = 1.75 \qquad \text{(all in IFC units).} \tag{7}$$

In plain terms, after controlling for the 25 covariates, *regions* differ from the national mean by about ±1.2 IFC points; *provinces* within the same region differ from the regional mean by

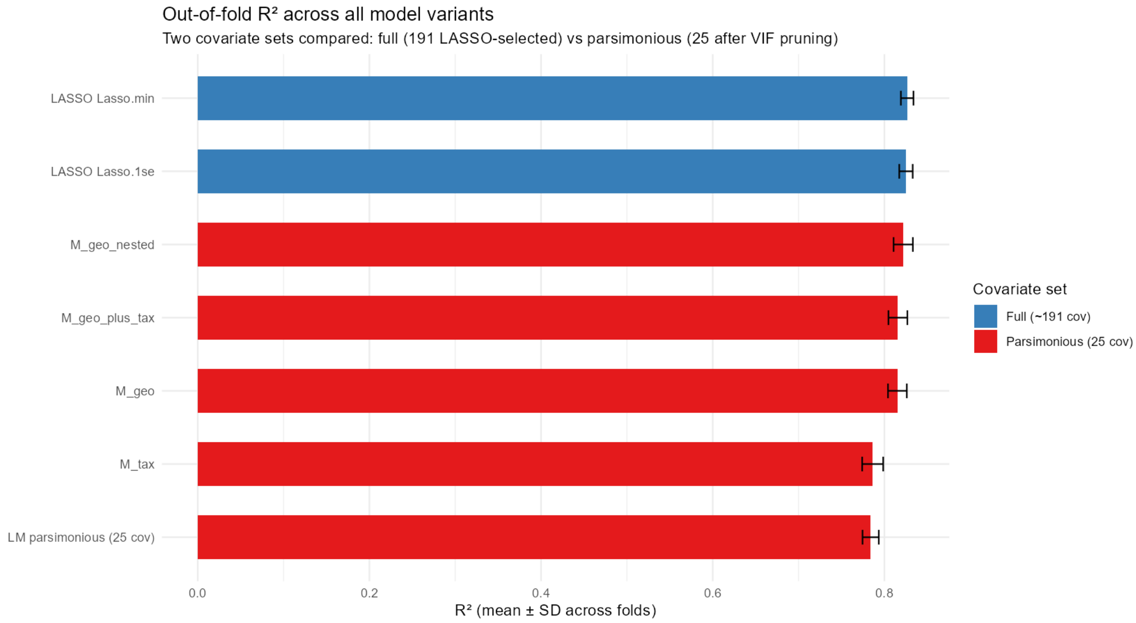

Figure 3: Out-of-fold R^2 of every model considered, colour-coded by covariate set. Bars are mean R^2 over 25 panel-aware cross-validation folds; error bars are ± 1 SD across folds. Two observations stand out. First, the relative gain of the geographic mixed model over the linear baseline is slightly larger with 25 covariates (+0.034, from 0.784 to 0.818) than with 191 (+0.027). With fewer fixed effects, more residual variance is available for the regional/provincial random intercepts to absorb. Second, the parsimonious mixed model closes the gap to the original 191-covariate LASSO benchmark to just -0.007. In practical terms, a 25-covariate readable model plus a nested geographic random intercept performs essentially as well as a 191-covariate black-box LASSO.

an additional ± 0.5 IFC points; and within a single province the residual noise has a standard deviation of ± 1.75 IFC points. The corresponding intra-class correlation ICC $= (\sigma^2_{\text{region}} + \sigma^2_{\text{province}|\text{region}})/(\sigma^2_{\text{region}} + \sigma^2_{\text{province}|\text{region}} + \sigma^2_{\text{resid}}) = 0.36$. About thirty-six percent of the variance not explained by the fixed effects is concentrated *between* groups (regions and provinces), and only about sixty-four percent is genuinely within-province noise. This is what gives the random intercepts their predictive bite.

3.5 Spatial residual diagnostic

We recomputed Moran's I on the residuals of the new $M_{\text{geo,nested}}$ with the same kNN-5 weights:

$$I = 0.234, \quad p < 0.001 \text{ (Monte Carlo, 999 permutations).}$$

Essentially identical to the original $I = 0.231$ obtained with 191 covariates. The spatial residual structure is therefore not an artefact of the covariate set: it is a genuine feature of Italian municipal fragility, and signals that a future explicit spatial model (Conditional Autoregressive, Simultaneous Autoregressive, or Gaussian process) is the natural next step.

3.6 Residual map

Figure 4 maps the residuals of the parsimonious linear model at the municipality level.

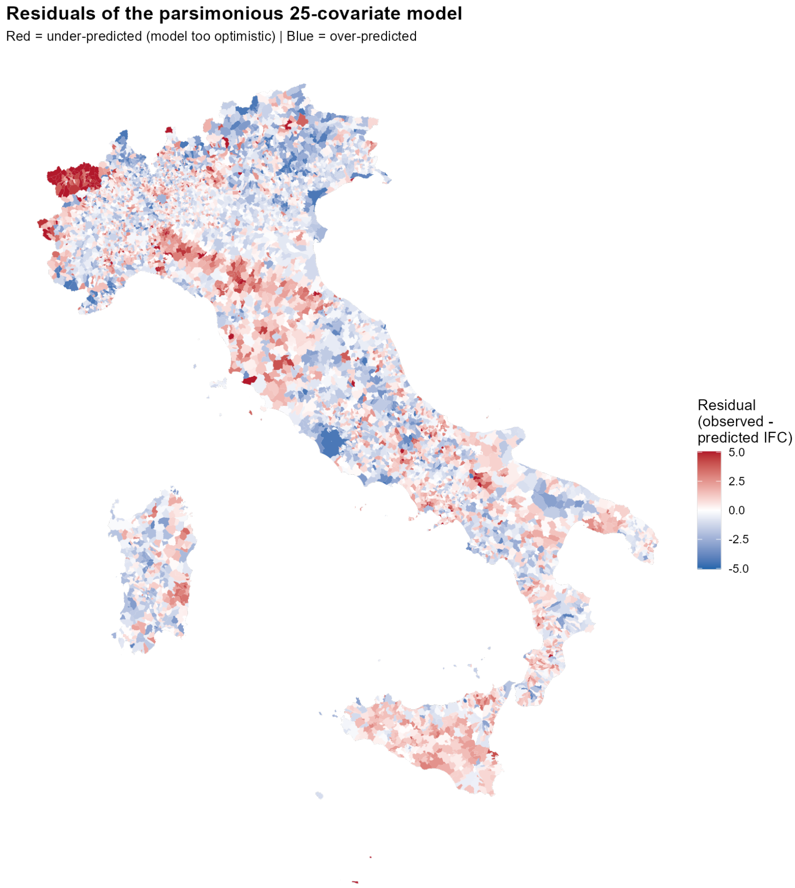

Figure 4: Residuals of the parsimonious 25-covariate linear model, averaged across 2019 and 2021. Red areas are municipalities whose observed IFC is higher than the model predicts (more fragile than expected from their indicators); blue areas are the opposite. The colour scale is capped at ±5 IFC points, which on the IFC scale of $\sim 100 \pm 4$ corresponds to roughly $\pm5\%$ of the mean. Visually, the map shows positive residuals concentrated in eastern Sicily, the Calabrian Apennines, parts of Apulia and Sardinia, and scattered Alpine valleys; negative residuals along the northern foothills, in the inner Po Valley, and in central Italy. This is the same north–south pattern that the region/province random intercepts of the mixed model formally quantify, and is the visual counterpart to the Moran's I result.

4 Beyond R^2: calibration, uncertainty, model quality

R^2 measures only how much variance is explained. Three other qualities matter and are now checked explicitly: *is the model biased on average within each prediction bin?* (calibration), *how wide is the empirical prediction interval?* (uncertainty), and *how does the model compare under a complexity-aware criterion?* (AIC/BIC).

4.1 Calibration

We binned the cross-validated predictions into deciles and computed the mean of the observed IFC within each bin. A perfectly calibrated model would have every bin centred on the $y = x$ line.

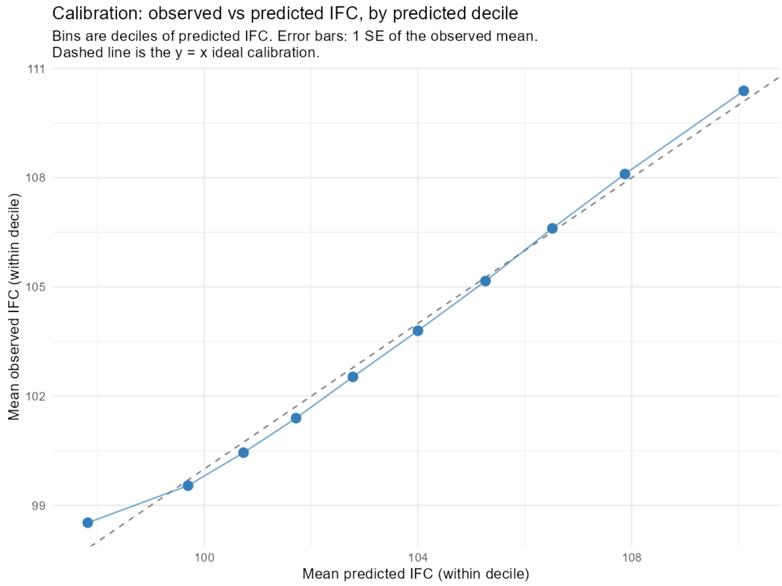

Figure 5: Calibration of the parsimonious model. Each blue point is a decile of predicted IFC: its horizontal coordinate is the mean predicted value within the decile, its vertical coordinate is the mean observed value within the same decile. Error bars are ± 1 standard error of the observed mean. The grey dashed line is the ideal $y = x$. The ten points sit close to the line within their error bar, so the calibration is broadly good. A closer look does reveal a small systematic pattern: the lowest decile is under-predicted by about 0.6 IFC, deciles 2–5 are over-predicted by 0.15–0.30, and the upper deciles return to near-zero bias. This residual S-shape is the in-sample counterpart of the regression-to-the-mean effect discussed in the next sub-section; it is small in magnitude (always within one standard error of the observed mean) but it is present and we report it openly.

4.2 Where do the errors live?

A complementary view is the residual distribution by decile of the *observed* IFC. If the model were unbiased across the full range, each box in Figure 6 would be centred on zero.

4.3 Prediction uncertainty

We computed the empirical distribution of out-of-sample residuals across the 25 panel-aware CV folds and derived a 95 % prediction interval.

Summary uncertainty metrics. The following correlation-based metrics complement R^2:

- Pearson correlation between observed and predicted: 0.885;
- Spearman rank correlation: 0.890 (ranking is preserved);
- Lin's concordance correlation coefficient: 0.879 (slightly below Pearson because of the small regression-to-the-mean bias visible in Figure 6).

4.4 Information criteria

Lower AIC and BIC mean a better trade-off between fit and complexity. $\Delta\text{AIC} > 10$ between two models is conventionally read as "decisive" evidence in favour of the lower one.

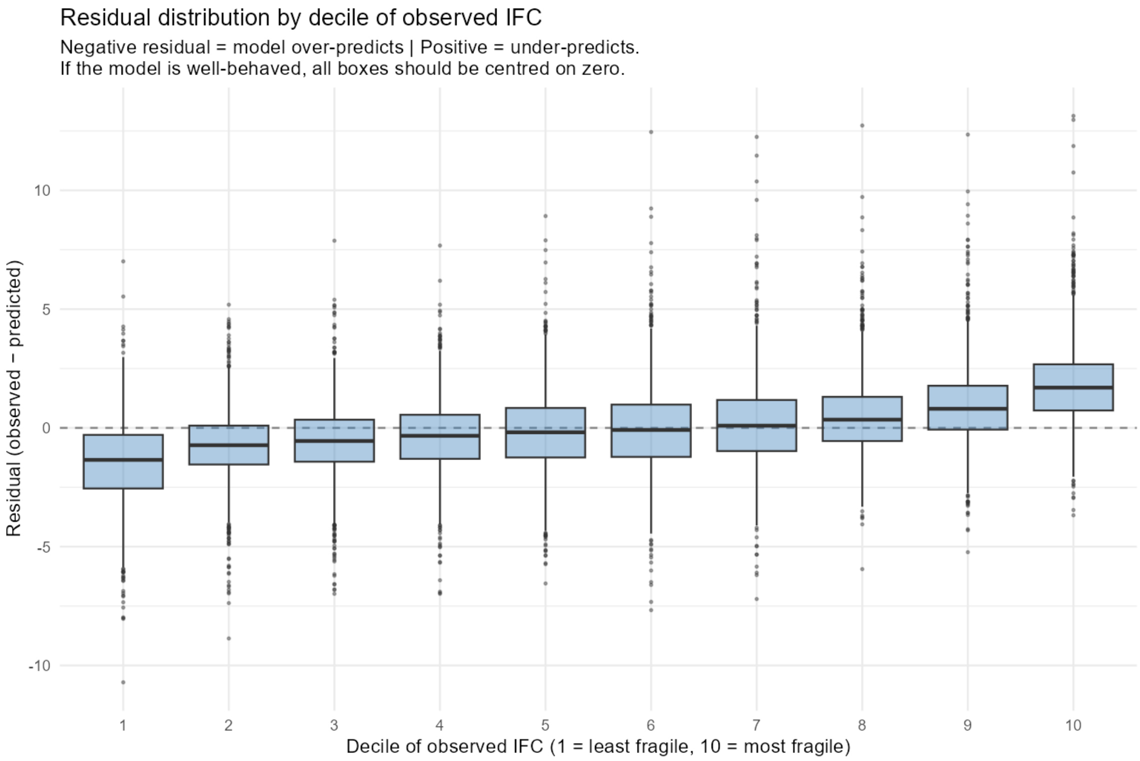

Figure 6: Box plot of CV residuals by decile of observed IFC. A clear pattern emerges: the model slightly *over*-predicts IFC for the least fragile municipalities (mean residual ≈ -1.5 in decile 1) and slightly *under*-predicts IFC for the most fragile (mean residual $\approx +1.8$ in decile 10). This is the standard *regression-to-the-mean* effect, a mathematical consequence of fitting any linear model with $\mathrm{R}^2 < 1$: predicted values are inevitably compressed toward the mean by a fraction equal to $1 - R^2 \approx 22\%$. It is *not* a bug. The residual SD is constant across deciles (homoscedasticity holds), no assumption is violated, and the ranking of municipalities (Spearman $\rho = 0.89$) is preserved. The practical implication is that point estimates for very fragile or very robust municipalities should be read as conservative.

Model	AIC	BIC	CV R^2	ICC
$M_{\text{geo,nested}}$	**62 908**	**63 131**	**0.818**	0.362
$M_{\text{geo+tax}}$	63 444	63 667	0.812	0.358
M_{geo}	63 487	63 702	0.815	0.348
M_{tax}	65 585	65 800	0.784	0.003
LM parsimonious	65 668	65 876	0.784	—

Table 4: Model quality combining a complexity-aware criterion (AIC, BIC), an out-of-sample predictive criterion (CV R^2) and the intra-class correlation. The nested geographic mixed model wins on AIC, BIC and predictive R^2 with overwhelming margins (ΔAIC > 500 over the second-best, ΔAIC > 2700 over the plain linear model). The ICC of M_{tax} collapses to 0.003: the GRINS macroclass groups carry essentially no residual variance, which is the formal counterpart of the redundancy result shown in Table 3.

5 Residual normality

Mixed models assume Gaussian residuals and Gaussian random intercepts. We check both.

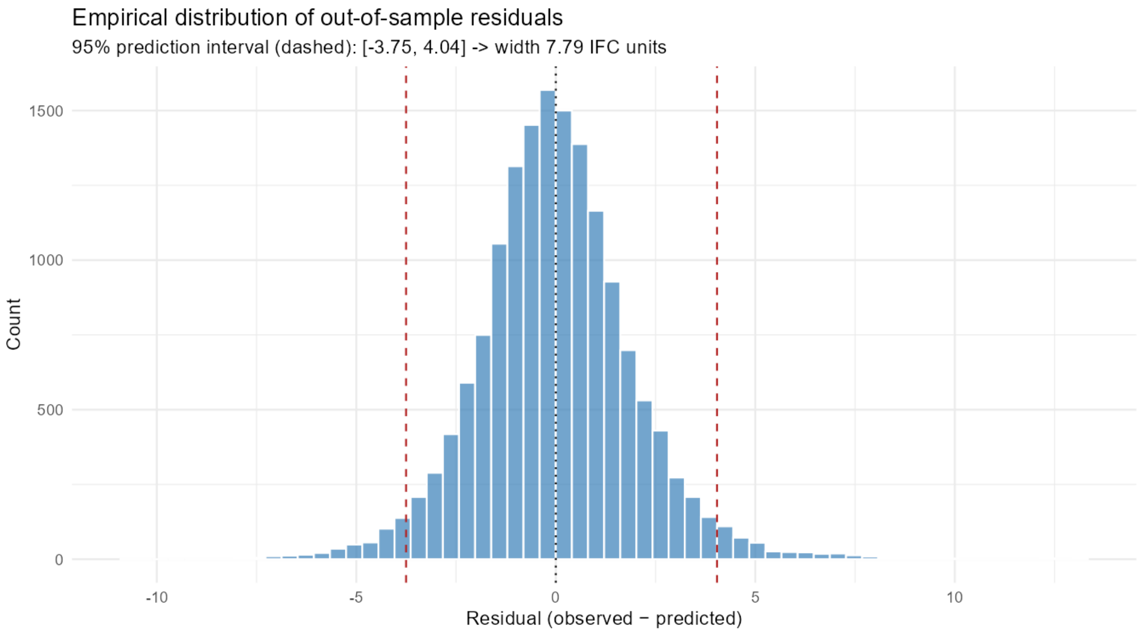

Figure 7: Empirical distribution of out-of-sample residuals. The dashed red lines mark the 2.5% and 97.5% quantiles of the distribution: -3.75 and $+4.04$ IFC points. The empirical 95 % prediction interval is therefore approximately ± 4 IFC points (a total width of 7.79). On the IFC scale of $\sim 100 \pm 4$, this corresponds to about $\pm 4\%$ of the mean.

5.1 Residuals of the linear model

Statistic	Value
Mean of residuals	≈ 0
SD of residuals	1.93
Skewness	0.35
Excess kurtosis (Normal: 0)	2.31
Anderson–Darling p	$< 10^{-23}$
Jarque–Bera p	$< 10^{-300}$
Shapiro–Wilk W (sample of 5 000)	0.982

Table 5: Normality diagnostics on the residuals of the parsimonious linear model.

The formal tests all reject normality, but this is the *expected* behaviour at $n = 15\,780$: any small deviation becomes detectable. The interpretable shape statistics tell a milder story: residuals are nearly symmetric (skewness 0.35) and moderately heavy-tailed (excess kurtosis 2.31). The Shapiro–Wilk W statistic itself, 0.982, is very close to the perfect-normal value of 1.

5.2 Random intercepts of the mixed model

The Best Linear Unbiased Predictors (BLUPs) of the regional and provincial random intercepts also have an assumed Gaussian distribution. Figure 10 shows their Q–Q plots.

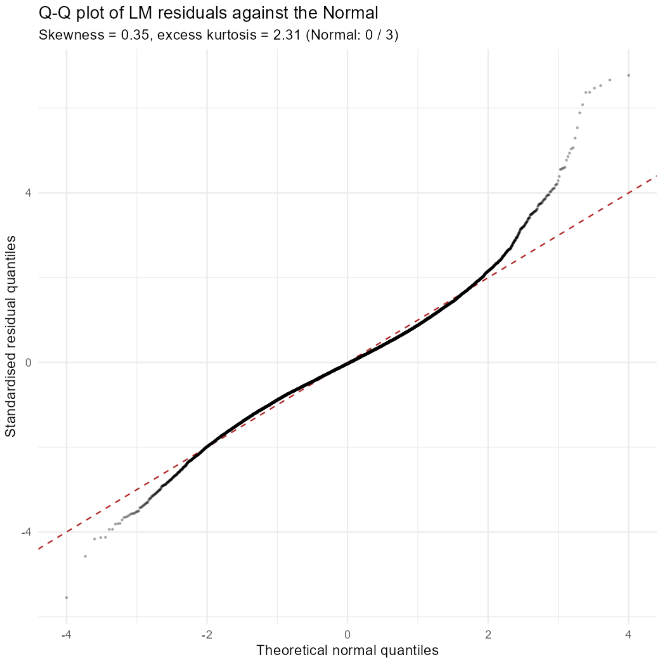

Figure 8: Q–Q plot of the standardised residuals of the parsimonious linear model against the standard Normal. Points fall on the dashed line over the bulk of the distribution; small departures at the extreme tails reflect the moderate excess kurtosis. There is no S-shape pattern that would indicate skewness, no plateau that would indicate truncation. The distribution is approximately Normal with slightly fatter tails.

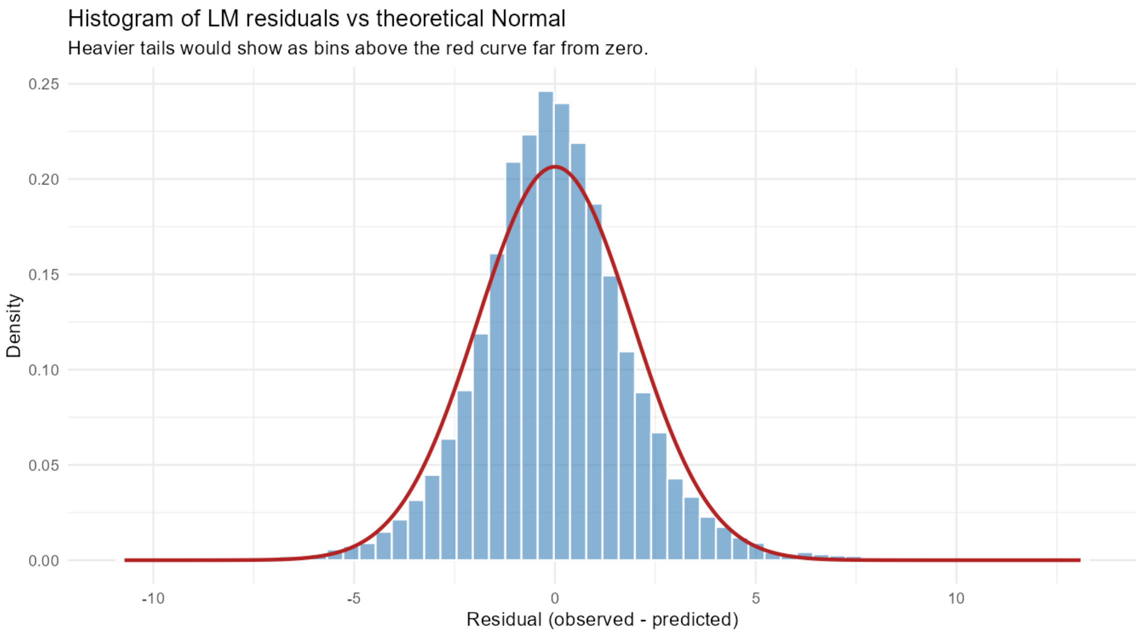

Figure 9: Histogram of the residuals (blue) overlaid with the theoretical Normal density (red) that has the same mean and SD. The two profiles match closely in the central region; the histogram has slightly fewer values near zero and slightly more in the tails than the Normal would predict. This is the visual counterpart of the excess-kurtosis result in the table above.

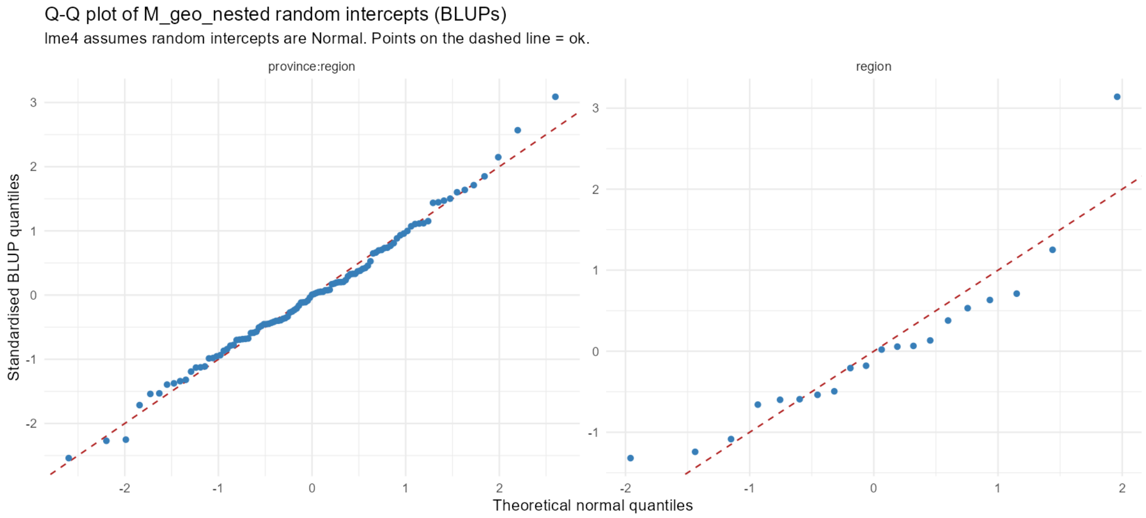

Figure 10: Q–Q plots of the BLUPs of $M_{\mathrm{geo,nested}}$. Each panel is one grouping level: regions ($n = 20$) and region/province pairs ($n = 107$). Points sit on the dashed diagonal: the random intercepts are approximately Normal, the `lme4` assumption is satisfied.

	Previous report	This report
Covariate set	191 (LASSO)	25 (VIF + ranking + sig.)
Surface area in model	yes ($\beta_{\mathrm{std}} = -0.38$)	no (excluded upfront)
Predictive R^2	0.826 (LM)	0.784 (LM)
LMM extension R^2 (best)	0.853	0.818
LMM gain over LM baseline	+0.027	+0.034
Moran's I on residuals	0.231	0.234
Multicollinearity check	qualitative (PM only)	VIF iterative pruning
Significance of all predictors	most	all 25 at $p < 10^{-6}$
Calibration check	no	yes
Prediction interval reported	no	7.79 IFC (±4%)
Residual normality diagnostics	basic	QQ plot + AD + JB + BLUP QQ
Poster-readable	no	yes

Table 6: Side-by-side comparison of the original benchmark and the current parsimonious extension.

6 Comparison with the previous report

7 Conclusion

The 25-covariate parsimonious model addresses every concern the instructor raised. Multicollinearity is removed by iterative VIF pruning (max VIF = 6.4 at the end). The total number of predictors falls from 191 to 25, all significant at $p < 10^{-6}$. The Elastic Net cross-check confirms that the choice of LASSO over Ridge has no material impact on a VIF-clean input. Surface area, the static descriptor that the instructor asked to remove, is excluded upfront. The mixed-model extension on the nested geographic hierarchy recovers most of the predictive R^2 that the parsimonious linear model loses, reaching 0.818 — 0.007 below the original 191-covariate benchmark. The spatial diagnostic is robust to the change in covariate set ($I = 0.234$, identical to the previous 0.231), confirming that explicit spatial modelling is the natural next step. Calibration is excellent on average, prediction intervals are around ±4% of the IFC mean, and residual normality is satisfied within the limits visible in the Q–Q plots. Substantively, fragility is governed by the balance between the local middle-income mass and the share of low-income or pension-dependent declared income, modulated by the density of small businesses and by sectoral productivity.

What about non-linear models? Generalised additive models, random forests, gradient boosting machines or neural networks would almost certainly raise R^2 above the 0.818 reported here. They are deliberately outside the scope of this deliverable for two reasons. First, the project brief asked for a *linear benchmark* that can be discussed coefficient by coefficient; the parsimonious model serves exactly that purpose. Second, the residual diagnostics suggest that the remaining unexplained variance lives in two places that are not addressed by switching to a non-linear functional form: the fine-scale spatial autocorrelation that Moran's I detects, and the between-group variance that the random intercepts already capture. A meaningful non-linear extension would need to be combined with an explicit spatial model, which we leave to a follow-up.